SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Kita Akan Lawan
Project / Solution Name	StockNow
Name (Leader)	Alfred Jason Gustono
Email & Phone (Leader)	alfredgustono0926@gmail.com / +62 821 6132 7646
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://stock-now-theta.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/Albrea/Kita-Akan-Lawan-StockNow.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Owners, Managers, and Staff use StockNow in a browser, languages in English, Bahasa Melayu, Tamil, or Chinese. One Next.js app on Vercel serves the React screens and the server routes. Every change goes through one server route that checks the user's role, applies the change and writes the activity log. Data is stored in Supabase PostgreSQL through supabase-js.

The forecast and restock engine is our own JavaScript code: it calculates forecasts, quantities, buy/wait/don't-buy decisions, profit, confidence and backtests. The Google Gemini API is called only from the server for three jobs: proposing forecast settings from derived statistics (validated, backtested and approved by an owner), writing the plan summary, and drafting supplier WhatsApp messages. It never decides what to buy. It falls back from gemini-3.5-flash to gemini-3.5-flash-lite to a built-in template.

Orbyte is an assumed interface: there is no live Orbyte access, so StockNow never calls Orbyte. We built StockNow's receiving side, a signed webhook that turns paid orders into stock-outs (off unless a secret is set), and a demo button that simulates Orbyte's payment. Other Orbyte modules are simulated. See Appendix A.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript on Node.js (v20+)	Language and runtime for the UI, server routes and engine. Node's built-in crypto signs/verifies the Orbyte webhook (HMAC-SHA256)	MIT (Node.js)
Next.js 15.5.28	Web framework: pages, server routes, production build	MIT
React, React DOM 19.2.0	User-interface components	MIT
lucide-react 0.555.0	Interface icons	ISC

Name & Version	Purpose in Project	License
@supabase/supabase-js 2.117.0	Reads and writes the Supabase PostgreSQL database	MIT
Vitest 3.2.7 (development only)	48 automated tests (engine, budget, approvals, undo, payments, Gemini fallback)	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (Google)	(1) Proposes forecast settings as JSON from derived sales statistics (2) writes the restock plan summary (3) drafts supplier order/reminder messages; plus a status check. Models: gemini-3.5-flash, backup gemini-3.5-flash-lite	Free tier
Supabase API (Supabase Inc.)	Database reads and writes for all app data	Free tier
Orbyte Payment, assumed interface (simulated, no live Orbyte access)	StockNow receives signed paid-order events at /api/orbyte/payment. No live Orbyte API is called. In the demo the events are simulated inside the app	None
WhatsApp click-to-chat, wa.me (Meta)	Opens a pre-filled supplier message; a person sends it. No API key, no automatic sending	Free
Google Fonts (Google)	Serves the three fonts above	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit, Inc.)	Hosted the Phase 1 prototype (Replit Deployments). Replaced by Vercel for the final version because publishing on Replit has an issue	Sponsor-provided (SDC) / Malaysia

Service & Provider	Purpose in Project	Plan / Region
Supabase (Supabase Inc.)	Managed PostgreSQL (11 tables, row-level security on): stock, sales, orders, logs, and each shop's business type and approved forecast settings. Schema in supabase/schema.sql. Existing databases run supabase/upgrade-final.sql once. No Supabase authentication used in the demo	Free plan / Malaysia
Vercel (Vercel Inc.)	Hosts and deploys the final Next.js app used for the demo and judging (https://stock-now-theta.vercel.app/); stores GEMINI_API_KEY, SUPABASE keys and ORBYTE_WEBHOOK_SECRET as encrypted server-side environment variables	Free (Hobby) plan / Vercel global network

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Google Gemini 3.5 Flash (primary)	Google, via Gemini API	Pre-trained, used as-is (no fine-tuning). JSON mode for settings proposals	Gemini API Additional Terms of Service
Google Gemini 3.5 Flash-Lite (automatic backup)	Google, via Gemini API	Pre-trained, used as-is, called only when 3.5 Flash is busy, slow or unavailable	Gemini API Additional Terms of Service (kenapa 2 sternt??)
StockNow forecast & restock engine (weighted moving average + linear trend, rule-based budget allocation, rolling backtest)	Built by the team	Self-built statistical rules, no model training. Six settings are adjustable within fixed ranges; Gemini may propose values, a person approves	Team-owned (Apache-2.0 repository)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
StockNow demo data (8 products, 6 months of sales, 6 demand scenarios, fictional staff and suppliers. Dates moved to the current day when loaded)	Created by the team; built into the app and the Supabase seed	Synthetic	Team-owned	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Backtest and benchmark results (forecast error per scenario, 2,000-product timing test)	Generated by the team from the synthetic data	Synthetic (evaluation)	Team-owned	N
DOSM MSMEs Performance 2024	dosm.gov.my (cited in the reports only, not loaded into the app)	Public	DOSM terms of use	N
SME interview notes and photos (KIPMall Kota Warisan, 5 shops)	Self-collected by the team; used only in the report appendix, never loaded into the app	Self-collected	Team-owned	Y (photos of shop staff). They consent to all.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic) Opus 5.5 Medium and High effort	• **Coding:** pages, the forecast engine and adaptive settings, Gemini integration, Orbyte payment webhook, undo, languages, database connection. • **Design:** layouts, light mode, phone version, loading screens. • **Testing and fixing:** help to fix tests case, browser runs, diagnosing the Gemini 3.5 timeouts and the deployment setup. • **Documents:** drafting and editing the README, pitch script and these Phase 2 documents; translating screen text into Malay, Tamil and Chinese. • **Research:** Gemini models and limits, Malaysian SME statistics.	Major
Replit Agent 4	Initial prototype, brainstorming, and deployment of early systems prototype. At Phase 2, Replit is used to refine the design to be more attractive and simple.	Moderate

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT (OpenAI) Free	Reviewing the project, preparing likely judge questions, and planning the adaptive forecasting feature and positioning (the plan was then implemented and tested)	Moderate
Vercel Version 59.16.0	Deployment Phase 2	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed AI-generated outputs by checking their accuracy, relevance, and consistency with project requirements. We conducted black-box exploratory testing by interacting with the system as end users, evaluating its functionality, usability, and feasibility without examining the underlying code. Any errors, misleading information, or unexpected behaviour were identified and corrected later on. The testing team discussed and verified the outputs before accepting them. We take full responsibility for the final deliverables, ensuring that AI was used as an assisting tool rather than a replacement for human judgement.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit cloud platform (hosting & deployment)	Built and hosted the Phase 1 prototype (Replit Agent and Replit Deployments). The final demo for judging runs on Vercel because publishing on Replit has an issue	Replit cloud platform (hosting & deployment)

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not essential to the core functionality. Replit hosted our Phase 1 prototype, but publishing has become an issue, so the final version used for the demo and judging runs on Vercel. StockNow is a standard Next.js application with its data in Supabase, so moving host mainly meant setting the same environment variables on Vercel. The forecast engine and Gemini integration are our own code and depend on neither Replit nor Vercel.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo uses synthetic data only, no real customer or personal data is collected. In real use StockNow stores business records (products, prices, stock movements, plans, orders, supplier business contacts) and staff names and roles for the activity log. No customer data is stored; an Orbyte payment event carries only a payment ID, SKUs and quantities.

Data is kept in Supabase PostgreSQL (encrypted at rest, HTTPS in transit, row-level security). Gemini receives only computed figures, product and supplier business names, or derived statistics (averages, variability, trend, margins); never raw sales records, phone numbers or staff details. Keys are Vercel server-side environment variables, webhook events must carry a valid HMAC signature.

Before real use we will add authentication (Orbyte sign-in or Supabase Auth). Records are kept while the business uses StockNow and can be exported or deleted on request, following Malaysia's PDPA 2010.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Forecast: needs at least three months of sales and models trend, so festive spikes are under-forecast (25.6% error in our festive scenario vs 7.8% overall). New shops start on business-type defaults. Mitigation: confidence score, "Show calculation", backtests, human approval of every item.

Adaptive settings: tuned and tested on the same past months, so improvements may be optimistic. Mitigation: fixed safe ranges, comparison with the current settings and a simple baseline, owner approval, one-click reset.

Gemini: can be wrong, slow or unavailable (3.5 Flash timed out in our tests). It never calculates, a number check rejects invented figures, and a backup model and template keep the app working.

Data and security: results are on synthetic data. Orbyte is simulated. The demo has no login and must not hold real data until authentication is added.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

8. Declaration

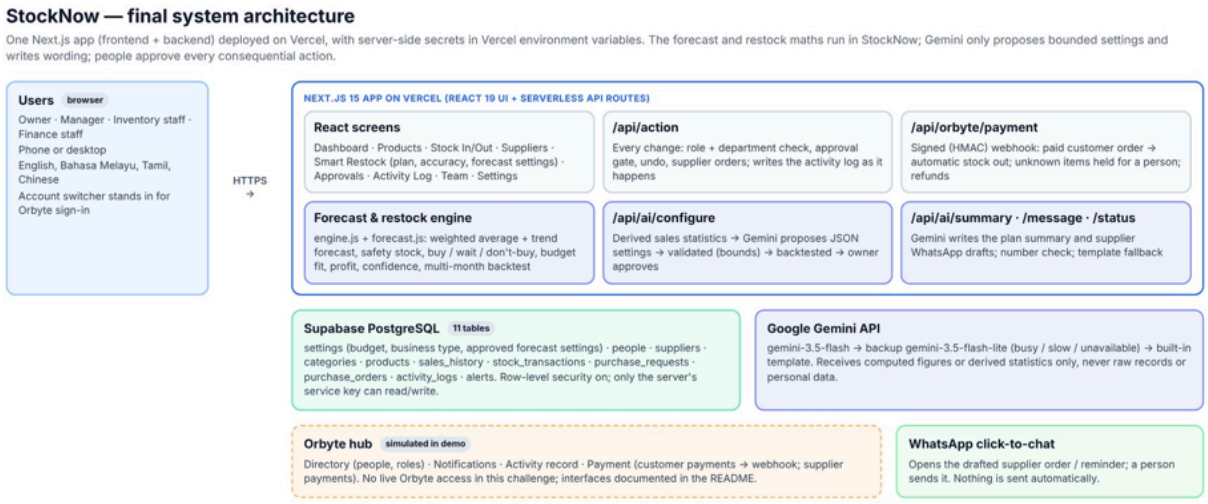
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Alfred Jason Gustono
31 October 2026

Appendix

Appendix A. System block diagram



StockNow final architecture: users → Next.js app on Replit (screens, API routes, forecast engine) → Supabase PostgreSQL and Google Gemini API; Orbyte (simulated) and WhatsApp click-to-chat.